

Did the new models meet expectations? — Phase 9 summary

CLAIM 1: FlexibleNN depth-regularisation (ELBO, DEPTH_PENALTY) learns to pick shallower architectures when the data allows.

- > MET post-fix. Both regulariser types drop mean depth vs NONE on simple data; COST_AWARE_ELBO goes further and is the MSE leader on B3 two-phase. (Earlier apparent 'success' before fixes was illusory — `n_predictor` was frozen.)

CLAIM 2: ProbReg's classification bottleneck helps on noisy regression.

- > PARTIAL. Ranked by MSE on the noise-robustness sweep:
 - sigma=0.05: NN > LGBM > XGB >> ProbReg dyn-k >> ClassReg
 - sigma=0.30: LGBM ~ NN ~ ProbReg dyn-k >> ClassReg
 - sigma=1.00: NN ~ ProbReg dyn-k (1.099) > ClassReg k=2 (1.162) > LGBMSo ClassReg's bottleneck DOES help vs tree baselines at high noise, but not vs a plain probabilistic NN. Paper narrative needs reframing.

CLAIM 3: Dynamic-k adapts per-input to the data.

- > MET only without ELBO. The ELBO prior pins `k=2` regardless of sigma. With `NClassesRegularization.NONE`, `mean_k` rises from ~SENTINEL (direct regression) at sigma=0.05 to `k~5` at sigma=1.0 — a clean adaptation story. Recommendation: report dyn-k results without the heavy ELBO prior, or retune prior strength.

CLAIM 4: SEPARATE_HEADS underperforms SINGLE_HEAD_FINAL on small data.

- > NOT MET at param-matched sizes. The earlier observation was driven by param-count mismatch (SEP has `n_classes` heads vs SINGLE's one head). Once parameter counts are aligned they are within noise on all `n_samples` in {200, 500, 1500}. SEP_HEADS retains the interpretability advantage (per-class regression heads) without a measurable small-data penalty.

CLAIM 5: Gumbel-Softmax + ELBO is broken.

- > NOT MET (previously believed). After N1/N2 + the optimizer-inclusion fix, Gumbel+ELBO trains and produces non-trivial selection entropy. It is still slightly worse than SoftGating+ELBO on MSE, but no longer a failure mode. Treat as a viable-but-lossier alternative, not broken.

OVERALL: multiple prior benchmark claims about FlexNN depth regularisation silently ran on frozen `n_predictor` weights. Those results need re-running. The new findings post-fix are encouraging but recalibrate the narrative against plain probabilistic NN baselines.

Verdict: PARTIAL: 2 MET, 2 NOT MET (re-framed), 1 partial